

Quan Huynh

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COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Chicago, Illinois 60607-7052

Quiz #1

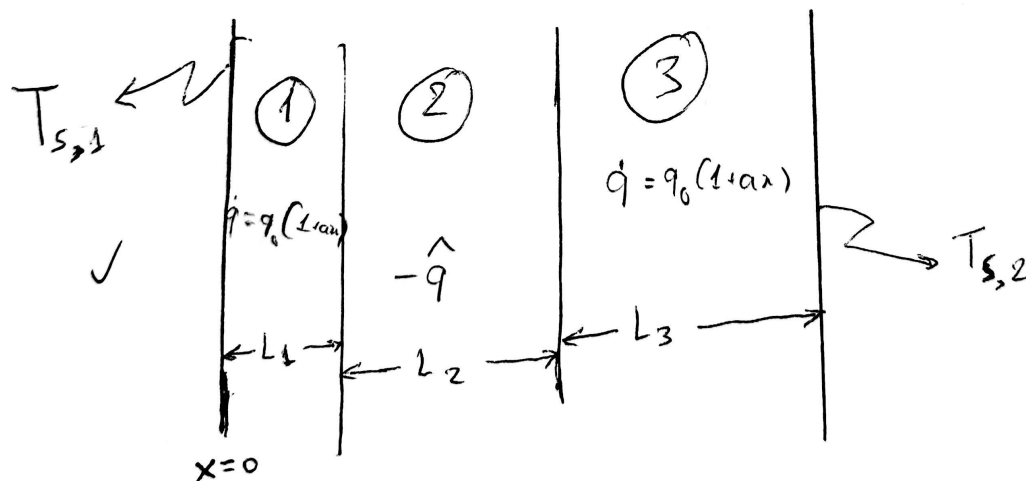
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Open Notes and Book

Duration: 40 minutes

Consider three sequential one-dimensional plane walls of thickness L_1 , L_2 and L_3 , respectively; heat is being transferred through this set of plane walls since the surface temperatures of the walls are maintained at $T_{s,1}$ and $T_{s,2}$ as shown in the sketch. The first and third ones experience $\dot{q} = \dot{q}_0(1 + a \cdot x)$ volumetric heat generation; the middle one involves uniform volumetric heat removal $\hat{q}$. All properties are constant.

- 1) Write all design/balance equations and corresponding boundary conditions
- 2) Determine the temperature profile $T(x)$ and heat flux distribution $q''(x)$.



1) Balance equations:

$$k_1 \frac{d^2 T_1}{dx^2} + \dot{q}_0 (1 + \alpha x) = 0$$

$$x \in [0, L_1]$$

$$k_2 \frac{d^2 T_2}{dx^2} - \hat{q} = 0$$

$$x \in [L_1, L_1 + L_2]$$

$$k_3 \frac{d^2 T_3}{dx^2} + \dot{q} (1 + \alpha x) = 0$$

$$x \in [L_1 + L_2, L_1 + L_2 + L_3]$$

Boundary Conditions

$$T_1|_{x=0} = T_{s,1} \quad \checkmark$$

$$T_3|_{x=L_1+L_2+L_3} = T_{s,2} \quad \checkmark$$

given

$$T_1|_{x=L_1} = T_2|_{x=L_1} \quad \checkmark$$

Continuity of T

$$T_2|_{x=L_2} = T_3|_{x=L_2} \quad \checkmark$$

$$k_1 \frac{dT_1}{dx} \Big|_{x=L_1} = k_2 \frac{dT_2}{dx} \Big|_{x=L_1} \quad \checkmark$$

Continuity of heat flux.

$$k_2 \frac{dT_2}{dx} \Big|_{x=L_2} = k_3 \frac{dT_3}{dx} \Big|_{x=L_2} \quad \checkmark$$

$$2) \quad k_1 \frac{d^2 T_1}{dx^2} = -\dot{q}_0 - \dot{q}_0 a x \Rightarrow \frac{d^2 T_1}{dx^2} = -\frac{\dot{q}_0}{k_1} - \frac{\dot{q}_0 a}{k_1} x$$

$$\frac{dT_1}{dx} = A - \frac{\dot{q}_0}{k_1} x - \frac{\dot{q}_0 a}{2k_1} x^2$$

$$T_1(x) = B - A x - \frac{\dot{q}_0}{2k_1} x^2 - \frac{\dot{q}_0 a}{6k_1} x^3 \quad (1)$$

B.C:

$$T_1(x=0) = T_{s,1} \Rightarrow T_{s,1} = B - 0 - 0 - 0 \Rightarrow \boxed{B = T_{s,1}}$$

$$k_2 \frac{d^2 T_2}{dx^2} - \hat{q} = 0 \Rightarrow \frac{d^2 T_2}{dx^2} = \frac{\hat{q}}{k_2} \Rightarrow T_2 = \frac{\hat{q}}{2k_2} x^2 + \overset{\text{Constants}}{C} x + D$$

B.C:

$$\left. \frac{dT_1}{dx} \right|_{x=L_1} = \left. \frac{dT_2}{dx} \right|_{x=L_1} \Rightarrow A - \frac{\dot{q}_0}{k_1} L_1 - \frac{\dot{q}_0 a}{2k_1} L_1^2 = \frac{\hat{q}}{k_2} L_1 + C$$

$$\Rightarrow \boxed{A = C + \frac{\hat{q}}{k_2} k_1 + \frac{\dot{q}_0}{k_1} L_1 + \frac{\dot{q}_0 a}{2k_1} L_1^2} \quad (1)$$

B.C:

$$T_1(x=L_1) = T_2(x=L_1) \Rightarrow \boxed{T_{s,1} - A L_1 - \frac{\dot{q}_0}{2k_1} L_1^2 - \frac{\dot{q}_0 a}{6k_1} L_1^3 = \frac{\hat{q}}{2k_2} L_1^2 + C L_1 + D} \quad (2)$$

$$k_3 \frac{d^2 T_3}{dx^2} = -\dot{q}_0 - \dot{q}_0 a x \Rightarrow \frac{d^2 T_3}{dx^2} = -\frac{\dot{q}_0}{k_3} - \frac{\dot{q}_0 a}{k_3} x$$

$$\frac{dT_3}{dx} = E - \frac{\dot{q}_0}{k_3} x - \frac{\dot{q}_0 a}{2k_3} x^2$$

$$T_3(x) = F + E x - \frac{\dot{q}_0}{2k_3} x^2 - \frac{\dot{q}_0 a}{6k_3} x^3$$

B.C:

$$T_3(x=L_1+L_2+L_3) = T_{s,2} \Rightarrow T_{s,2} = \boxed{F + E(L_1+L_2+L_3) - \frac{\dot{q}_0}{2k_3} (L_1+L_2+L_3)^2 - \frac{\dot{q}_0 a}{6k_3} (L_1+L_2+L_3)^3} \quad (3)$$

$$B.C: T_2(x=L_1+L_2) = T_3(x=L_1+L_2)$$

$$\frac{\hat{q}}{2k_2} (L_1+L_2)^2 + C(L_1+L_2) + D = F + E(L_1+L_2) - \frac{\dot{q}_0}{2k_1} (L_1+L_2)^2 - \frac{\dot{q}_0 a}{6k_3} (L_1+L_2)^3 \quad (4)$$

$$B.C: \left. \frac{dT_2}{dx} \right|_{x=L_1+L_2} = \left. \frac{dT_3}{dx} \right|_{x=L_1+L_2}$$

$$\frac{\hat{q}}{k_2} (L_1+L_2) + C = E - \frac{\dot{q}_0}{k_3} (L_1+L_2) - \frac{\dot{q}_0 a}{2k_3} (L_1+L_2)^2 \quad (5)$$

unk: A, C, D, E, F

Solve system of equation (1), (2), (3), (4), (5) for A, C, D, E, F